The Polynomial Compatibility Conjecture

Low-degree, low-influence polynomial distributions over an Abelian group

Status. Statement: AI-written from Per Austrin, Hao Chung, Kai-Min Chung, Shiuan Fu, Yao-Ting Lin, Mohammad Mahmoody, *On the Impossibility of Key Agreements from Quantum Random Oracles*, Cryptology ePrint Archive 2022, not yet checked by a human. The case of exponentially small influences, $\delta < |\mathcal{Y}|^{-d}/d$, is settled by the source (Theorem 4.4) and its counterexamples rule out $\delta \geq 1/(2d)$; the whole range in between — in particular the inverse-polynomial regime asserted here — is open.

Category. *Idealized Models & Non-Uniformity.*

Conjecture (informal). *Take two probability distributions over real-valued functions on a grid, each function built from a fixed small number of coordinates at a time (degree at most d), each normalised so that its average squared value is one, and each coordinate carrying only a tiny share of the total weight on average (influence at most δ). The conjecture says that if δ is small but still only inverse-polynomially small in d, then you can always draw one function from the first distribution and one from the second and find a single input at which both are nonzero. This is a purely Fourier-analytic statement, but it was isolated because it is exactly what is needed to break key agreement in the quantum random oracle model: the two distributions describe what the oracle looks like conditioned on the shared key being zero and on it being one, a common nonzero point is a single oracle consistent with both, and perfect agreement on the key then makes those two executions contradict each other. The authors could not prove it. They can prove it only when the influence bound is exponentially small in the degree, which still gives an attack, but an exponential-query one. They also give explicit counterexamples showing the influence bound cannot be relaxed beyond roughly one over twice the degree, and that dropping either the degree bound or the influence bound on either of the two distributions makes the statement false.*

1 The setting

Merkle's puzzles [Mer74] let two parties who share only a random oracle agree on a key using d queries while an eavesdropper needs $\Omega(d^2)$ queries, and a long line of work [IR89, BMG09, BM17] showed that this quadratic gap is the best possible in the classical random oracle model. Quantum access changes the picture: Grover search breaks Merkle's protocol with $O(d)$ queries, but quantum honest parties can restore a super-linear gap [BS08]. Whether parties who compute quantumly but communicate classically — the QCCC model — can achieve *super-polynomial* security in the quantum random oracle model was asked by Hosoyamada and Yamakawa [HY20] and is the motivating question of the source paper.

The paper reduces that question, for protocols with perfect completeness, to a statement in Fourier analysis. Following Zhandry's way of recording quantum queries [Zha19], one writes the oracle register in the Fourier basis; after the two parties make d quantum queries in total, the joint state of the parties and the oracle is d-sparse there, which is the same thing as saying that the associated "state polynomial" on the oracle's truth table has degree at most d and ℓ_2 norm 1. A query that the eavesdropper has not yet learned is one whose variable has small influence, so after an eavesdropper has learned all ε-heavy queries she is left with two distributions of exactly the kind in the conjecture: one for the residual state conditioned on the key being 0, one conditioned on the key being 1. A common point at which a polynomial from each is nonzero is a single oracle consistent with both executions, and perfect completeness then says the two executions cannot produce different keys. So the conjecture is precisely the missing step.

Granting it, the paper obtains a poly$(d, \log|\mathcal{Y}|)$-query *classical* attack on every perfectly complete QCCC key agreement protocol in the QROM, and, as a corollary, a conditional quantum black-box separation of QCCC key agreement from one-way functions, extending it to primitives such as oblivious transfer that imply key agreement in a black-box way. Unconditionally, the paper proves the conjecture only in the regime of exponentially small influences, $\delta < |\mathcal{Y}|^{-d}/d$, which yields an attack of $2^{O(dm)} \cdot d^2$ queries for oracles with m-bit outputs. Its counterexamples show the hypotheses cannot be dropped: both distributions need the degree bound, both need the influence bound, and δ must be below $1/(2d)$.

The statement is a cousin of the Aaronson–Ambainis conjecture [AA14], whose contrapositive also asserts that a low-degree polynomial with polynomially small influences must be large somewhere. Both are known when the influences are exponentially small [DFKO06], and the degenerate case of no communication and perfect completeness is settled by [OSSS05]; but the paper reports that the two conjectures do not appear to be directly comparable, and that their cryptographic consequences are incomparable too.

Progress to date. Theorem 4.4 proves the conjecture for $\delta < |\mathcal{Y}|^{-d}/d$ by fixing a maximum-degree term of f and showing that a g with small influences has, in expectation over a random assignment to the remaining variables, a constant term dominating all $|\mathcal{Y}|^d$ non-constant terms of the restriction; the $|\mathcal{Y}|^{-d}$ loss comes from that count (in the Boolean case $\mathcal{Y} = \mathbb{Z}_2$ this is the 2^{-d} of the paper's overview). The proof in fact uses no influence condition on F and no degree bound on G. Theorem 5.6 (Appendix A) shows it suffices to treat real-valued rather than complex-valued functions. Lemma 4.8 (Section 8) shows that the conjecture for one constant-size Abelian group already gives $\mathrm{poly}(d, \log|\mathcal{Y}'|)$-query attacks for oracles with range any finite Abelian group $\mathcal{Y}'$, so the choice of group in the conjecture is not the obstacle.

2 Notation and parameters

Throughout, $\mathcal{Y}$ is a finite Abelian group, written additively, and $\hat{\mathcal{Y}}$ is its dual group, i.e. the set of characters $\chi : \mathcal{Y} \to \{\omega \in \mathbb{C} : |\omega| = 1\}$ satisfying $\chi(y + y') = \chi(y)\chi(y')$. The trivial character, $\chi(y) = 1$ for all $y \in \mathcal{Y}$, is denoted $\hat{0}$. One has $|\hat{\mathcal{Y}}| = |\mathcal{Y}|$.

For $N \in \mathbb{N}$ write $[N] = \{1, \ldots, N\}$. A tuple $\chi = (\chi_1, \ldots, \chi_N) \in \hat{\mathcal{Y}}^N$ is viewed as the function $\chi(x) = \prod_{i=1}^{N} \chi_i(x_i)$ on $x = (x_1, \ldots, x_N) \in \mathcal{Y}^N$. These $|\mathcal{Y}|^N$ functions form an orthonormal basis for the complex-valued functions on $\mathcal{Y}^N$ under the inner product $\langle f, g \rangle = \mathbb{E}_x[f(x)\overline{g(x)}]$, where x is uniform on $\mathcal{Y}^N$. Consequently every $f : \mathcal{Y}^N \to \mathbb{R}$ has a unique Fourier expansion

$$f(x) = \sum_{\chi \in \hat{\mathcal{Y}}^N} \hat{f}(\chi)\chi(x), \qquad \hat{f}(\chi) \in \mathbb{C}.$$

The ℓ_2 norm is

$$\|f\|_2 \;=\; \left(\mathbb{E}_x[|f(x)|^2]\right)^{1/2} \;=\; \left(\sum_{\chi} |\hat{f}(\chi)|^2\right)^{1/2},$$

again with x uniform on $\mathcal{Y}^N$.

For a probability distribution F over functions $\mathcal{Y}^N \to \mathbb{R}$, $\mathrm{supp}(F)$ denotes its support, and $f \leftarrow F$ denotes a sample. The parameters are the degree bound $d \in \mathbb{N}$, the number of variables $N \in \mathbb{N}$, and the influence bound δ.

- $\mathcal{Y}$: a finite Abelian group; the range of the random oracle in the cryptographic application. The conjecture asserts that some such group works; the Boolean case $\mathcal{Y} = \mathbb{Z}_2$ is the special case stated as Conjecture 1.2 in the paper.
- N: number of variables, i.e. the size of the oracle's domain. Universally quantified; the influence bound must not depend on it.
- d: degree bound on the polynomials; in the application, the total number of quantum oracle queries made by the two honest parties.
- $\delta(d)$: bound on the average influence of each variable. The conjecture requires $\delta(d)$ to be at least inverse polynomial in d; the paper's counterexamples force $\delta(d) < 1/(2d)$.
- F, G: the two distributions over polynomials. In the application they describe the oracle's state conditioned on the agreed key bit being 0 and being 1 respectively.

3 The conjecture

Definition 1 (degree and influence). For $\chi = (\chi_1, \ldots, \chi_N) \in \hat{\mathcal{Y}}^N$ let $\deg(\chi) = \left|\{i \in [N] : \chi_i \neq \hat{0}\}\right|$. The *degree* of $f : \mathcal{Y}^N \to \mathbb{R}$ is

$$\deg(f) \;=\; \max\{\deg(\chi) \;:\; \hat{f}(\chi) \neq 0\},$$

with $\deg(f) = 0$ for f constant. For $i \in [N]$, the *influence* of the i-th variable on f is

$$\mathrm{Inf}_i(f) = \sum_{\chi : \chi_i \neq \hat{0}} \left|\hat{f}(\chi)\right|^2.$$

For $\mathcal{Y} = \mathbb{Z}_2$, identifying $\mathcal{Y}^N$ with $\{\pm 1\}^N$, these are the usual notions for multilinear polynomials: writing $f = \sum_{S \subseteq [N]} \alpha_S \prod_{i \in S} x_i$ with $\alpha_S \in \mathbb{R}$, one has

$$\deg(f) = \max_{\alpha_S \neq 0} |S|, \qquad \mathrm{Inf}_i(f) = \sum_{S \ni i} \alpha_S^2, \qquad \|f\|_2^2 = \sum_S \alpha_S^2.$$

Conjecture 1 (polynomial compatibility). *There exist a finite Abelian group $\mathcal{Y}$, a constant $c > 0$, and a function $\delta \colon \mathbb{N} \to (0,1]$ with $\delta(d) \geq d^{-c}$ for all $d \in \mathbb{N}$, such that the following holds for all $d, N \in \mathbb{N}$. Let F and G be probability distributions over functions $\mathcal{Y}^N \to \mathbb{R}$ such that:*

- unit ℓ_2 norm: $\|f\|_2 = 1$ *for every* $f \in \mathrm{supp}(F)$ *and* $\|g\|_2 = 1$ *for every* $g \in \mathrm{supp}(G)$*;*
- degree at most d: $\deg(f) \leq d$ *for every* $f \in \mathrm{supp}(F)$ *and* $\deg(g) \leq d$ *for every* $g \in \mathrm{supp}(G)$*;*
- δ-influences on average: *for every* $i \in [N]$,

$$\mathbb{E}_{f \leftarrow F}\left[\mathrm{Inf}_i(f)\right] \leq \delta(d) \quad \textit{and}$$
$$\mathbb{E}_{g \leftarrow G}\left[\mathrm{Inf}_i(g)\right] \leq \delta(d).$$

Then there exist $f \in \mathrm{supp}(F)$, $g \in \mathrm{supp}(G)$ and $x \in \mathcal{Y}^N$ such that $f(x) \cdot g(x) \neq 0$.

Remark 1 (which group, and how it is quantified). Conjecture 1 is the *existential* form: it asks only that some finite Abelian group $\mathcal{Y}$ admit an inverse-polynomial δ. This is the form taken by the source's Conjectures 4.3 and 5.5, and it is the form its Theorems 4.5 and 6.3 use; by Lemma 4.8 one constant-size group already suffices for attacks against oracles whose range is any finite Abelian group, so the choice of group is not where the difficulty lies. The paper's introductory Conjecture 1.2 fixes the Boolean case $\mathcal{Y} = \mathbb{Z}_2$; that version is strictly stronger and also open, and it must not be presented as the paper's main conjecture.

Remark 2 (how to read the hypotheses). Three points are easy to get wrong. First, the influence bound is on the *average* over the distribution, $\mathbb{E}_{f\leftarrow F}[\mathrm{Inf}_i(f)]$, and not on each polynomial in the support; imposing it pointwise strengthens the hypothesis and so weakens the conjecture. Second, N is universally quantified and δ depends on d alone: the source's Conjecture 5.5 leaves N free in the text, and the universal reading is the one made explicit in the equivalent Conjecture 4.3 ("for any $d, N \in \mathbb{N}$"). Third, the paper's introduction writes the ℓ_2 norm without the square root; since the constraint is that the norm equals 1 this makes no difference, and the standard normalisation of Section 2 is used here.

Remark 3 (the window that is open). The source proves Conjecture 1 (Theorem 4.4) whenever $\delta < |\mathcal{Y}|^{-d}/d$, i.e. for exponentially small influences, and shows (Theorem 5.6) that this polynomial formulation is equivalent to the quantum-state formulation, its Conjecture 4.3. Appendix B rules out $\delta \geq 1/(2d)$ and shows that the degree and influence hypotheses are needed on both distributions. Everything between $|\mathcal{Y}|^{-d}/d$ and $1/(2d)$ is open, and any δ that is $1/\mathrm{poly}(d, \log|\mathcal{Y}|)$ already suffices for a polynomial-query attack.

Remark 4 (what settling it would give). The conjecture is the whole gap between what is known and a full answer to the question of whether quantum parties talking over a classical channel can get super-polynomial security from a random oracle, at least in the perfectly complete case. Proving it gives an unconditional polynomial-query classical attack on every such protocol and a quantum black-box separation of QCCC key agreement — hence oblivious transfer and public-key encryption with classical inputs and communication — from one-way functions. Refuting it would not immediately give a secure protocol, but it would kill the only known route to one of the two main barriers in this area, and it would be a genuinely new phenomenon in the Fourier analysis of low-influence low-degree functions, a regime where the analogous Aaronson–Ambainis conjecture [AA14] is widely believed.

4 Bibliography

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